

RITICK SETHI

Sensor & Embedded Systems • Automation & Robotics • Solution Engineering •
Python & Data Tooling • Project Delivery • Technical Consulting
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PROFESSIONAL SUMMARY

Engineer with experience spanning sensor systems, embedded development, robotics, industrial automation, and customer-facing solution engineering. M.Sc. in Sensor Systems Technology with hands-on delivery across BMW, Bosch, FZI, FEV, and EP Equipment — covering Python tooling and AI prototyping, ROS2 robotics, V2G energy-system simulation, and on-site deployment of automated intralogistics. Technical consultant who builds and executes — diagnoses operational problems, designs solutions, writes production code, and delivers on-site. Operates across the full stack from strategic problem framing to AGV hardware troubleshooting in the field. Proven ownership pattern across BMW, FZI, FEV, and EP Equipment: every system started with a problem diagnosis and was delivered end-to-end.

CORE COMPETENCIES

Domains: Sensor Systems, Embedded & IoT, Robotics & Automation, Warehouse & Intralogistics, Energy Systems & V2G, Industrial Software Integration

Engineering: Solution & Project Engineering, Customer Requirement Analysis, System Architecture, On-site Commissioning, Stakeholder Management

Programming: Python (advanced), FastAPI, REST APIs, SQL (basic), JavaScript/React, HTML/CSS, C (basics)

Tools & Frameworks: Git, Postman, Docker (basics), CI/CD, Jira, Confluence, Agile/Scrum, Streamlit

Robotics & Simulation: ROS2, MuJoCo, Gazebo, Linux, Microcontrollers, AutoCAD (basics)

Data, ML & AI: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Power BI, Tableau, Ollama, BMW GAIA

Energy & Optimisation: PyPSA, oemof.solph, LP/MILP, V2G/PV/BESS modelling

Languages: English – C1 Fluent • German – B1 (actively improving) • Hindi – Native

PROFESSIONAL EXPERIENCE

Project Engineer – Automation | EP Equipment GmbH

Dec 2025 – Present • Cologne, Germany

- ▶ **Translated** customer requirements into structured automation solutions — material flow design, data flow architecture, AMR/AGV sizing, technical proposals, and quotations.
- ▶ **Identified** a structural bottleneck in the sales process — engineering was required in every customer meeting to extract viable requirements. Designed and built a Python/Streamlit self-service tool encoding solution logic, efficiency calculations, and quotation generation — removing the dependency entirely and reducing scoping time by ~40%. Self-initiated, not assigned.
- ▶ **Tested** WMS and fleet manager API integrations using Postman; identified discrepancies and improved internal integration documentation.
- ▶ **Coordinated** across customers, WMS vendors, AGV integrators, and internal fleet management teams — managing timelines, procurement sequencing, delivery risk, and weekly steering meetings without formal authority over any party.
- ▶ **Designed** and deployed live operational dashboards (HTML, CSS, JavaScript, Vue.js) on customer sites for real-time monitoring including FIFO logic and storage management.
- ▶ **Authored** standardized product specification documents, customer implementation manuals, and reusable layout/machine requirement appendices used by the business unit as the foundation for future projects and AI-supported sales tools.

Master Thesis – Vehicle-to-Grid Simulation | FEV Europe GmbH

Apr 2025 – Oct 2025 • Aachen, Germany

- ▶ **Developed** a modular Python simulation framework (oemof.solph, LP/MILP) to optimise V2G, PV, BESS, and grid-integrated systems — reduced simulated energy costs by 23%.
- ▶ **Built** an interactive Streamlit application and automated data preprocessing workflows (Pandas, NumPy) for scalable energy analysis.
- ▶ **Designed visual** dashboards (Matplotlib, Plotly) to evaluate fleet charging, load shifting, and cost-optimisation strategies. Thesis graded 1.1 (Sehr Gut).

Intern – Digital Platform Engineering (DevOps) | BMW AG

Oct 2024 – Apr 2025 • Munich, Germany

- ▶ **Developed** a Python-based tool to automate SIT result analysis and integrated data via REST APIs into ATC Jira — eliminated ~8 hrs/week of manual reporting.
- ▶ **Explored** BMW's internal AI tools (GitHub Copilot, GAIA, Dynatrace AI) and built a prototype chatbot for departmental knowledge support.
- ▶ **Contributed** to agile ceremonies, created user stories, and maintained documentation in Jira/Confluence; supported international teams with user access management in PTC Windchill PLM.

Working Student – IT Support | Robert Bosch GmbH

May 2024 – Oct 2024 • Karlsruhe, Germany

- ▶ **Resolved** hardware, software, and OS-related issues to support business continuity and onboarding of new staff.
- ▶ **Collaborated** with departments to deliver tailored IT solutions and small process improvements.

Student Research Assistant – Robotics | FZI Forschungszentrum Informatik

Feb 2024 – Apr 2024 • Karlsruhe, Germany

- ▶ **Contributed** a 3D motion planner for robotic climbing manoeuvres using ROS2 and MuJoCo; integrated ML-based navigation to improve agility.
- ▶ **Built** and deployed the front-end operator interface using HTML, CSS, JavaScript, and React.

Student Research Assistant – Sensor Systems | Karlsruhe Institute of Technology (KIT)

Aug 2023 – Dec 2023 • Karlsruhe, Germany

- ▶ **Reviewed** and validated optical and noise sensors for research applications; designed and executed sensor-based experiments with documented quality protocols.

EDUCATION

M.Sc. Sensor Systems Technology | Hochschule Karlsruhe (HKA)

Mar 2023 – Oct 2025

Thesis: Python-based Simulation Framework for V2G Integration in Multi-Source Energy Systems (FEV Europe GmbH) — graded 1.1 (Sehr Gut). **Focus areas:** Sensor Signal Processing, Embedded & Computer-Aided Sensorics, System Integration (1.0), Machine Learning, Management.

B.Tech. Electronics & Communication Engineering (IoT Specialisation) | Manav Rachna IIRS, India

Mar 2018 – Jul 2022

CGPA: 7.54 / 10. Final-year project on IoT-based virus-safe chamber (Vajra Kaksh); multiple publications and innovation awards.

PROJECTS

- ▶ **LinkedIn AI OS** — FastAPI + Claude/Ollama. Profile KPI scoring, RAG-based post generation, market analysis, 30+ routes, SQLite. github.com/Riticksethi6/linkedin-profile-optimizer
- ▶ **Warehouse Automation Pre-Sales Tool** — Customer input → material flow → AMR/AGV recommendation → WMS spec. Used live in EP Equipment sales cycles. github.com/Riticksethi6/warehouse-automation-presales-tool
- ▶ **V2G Energy Optimisation Framework** — PyPSA & oemof MILP simulation, 10+ scenarios, 50+ vehicle fleet, Streamlit dashboard. Master thesis graded 1.1.
- ▶ **Machine Learning Laboratory [MSc 2024]:** Explored ML techniques (EDA, PCA, regression, classification) using Python, scikit-learn, and real-world datasets.
- ▶ **Vajra Kaksh – IoT Virus-Safe Chamber** — Raspberry Pi with sensor fusion for real-time virus risk detection. Microsoft Imagine Cup 2021 Asia Semifinalist.

PUBLICATIONS, RECOGNITION & CERTIFICATIONS

Publications: Systematic Review of AI (IJRTI, 2022) • Vajra Kaksh (MR IJET, 2023) • Computer Vision (IEEE, 2020) • ML Applications (CEJ, 2020)

Awards: Best Innovative Idea – Innoskill 2022 • Asia Semifinalist – Microsoft Imagine Cup 2021 • 2nd – YASH 2020 • Semifinalist – IICDC TI 2019 • 2nd – Innoskill 2019

Certifications: Data Analyst Specialisation (Power BI, SQL, Python, R, Excel, Tableau) – Coursera • Cloud Computing with AWS – Udemy • IoT with Programming (Arduino, Raspberry Pi, Python) – edX • RPA Developer Foundation – UiPath